

Machine Learning for Cross-Sectional Stock Return Prediction

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1 Introduction and Problem Statement

Predicting stock returns is a central problem in quantitative finance, but it is also a difficult one because realized returns are noisy, unstable over time, and only weakly related to observable firm characteristics. In this project, we study whether machine learning models can extract useful cross-sectional signals from structured financial data and translate them into economically meaningful investment strategies.

The objective is to predict one-month-ahead stock returns for a broad panel of U.S. equities. Rather than focusing on the exact numerical value of future returns, we mainly aim to rank stocks within each month. This ranking perspective is more aligned with portfolio construction, where the relevant question is whether the model can identify stocks that are likely to outperform or underperform their peers.

We build a monthly stock-level panel combining market returns, accounting information, and empirical asset-pricing predictors. Several model classes are compared. We first define a small benchmark set, including a naive momentum signal, an XGBoost classifier, and a passive market benchmark based on the S&P 500 ETF. We then consider static deep learning models, namely an MLP and an FT-Transformer, which allow for nonlinear interactions between tabular predictors. Finally, we study a temporal tabular model designed to use sequences of past observations rather than only a single monthly feature vector.

The main research question is therefore:

Can machine learning models improve the cross-sectional ranking of future stock returns and generate useful portfolio signals out of sample?

To answer this question, we evaluate models using both forecasting and portfolio-based metrics. This allows us to distinguish statistical predictability from actual investment usefulness.

2 Data and Features

2.1 Data Sources and Panel Construction

We build a monthly stock-level panel from three structured financial datasets. CRSP monthly returns define the investment universe and provide the one-month-ahead return target. Compustat accounting data are linked to CRSP through the CRSP/Compustat Merged (CCM) link table and provide firm fundamentals. Finally, Jensen–Kelly–Pedersen (JKP) factor data are used to capture market-wide factor states related to value, momentum, profitability, investment, risk, trading activity, and liquidity [5, 6].

The raw CRSP file is first cleaned so that each observation is uniquely identified by a PERMNO and a month. Duplicate stock-month rows are removed or resolved only when the core return fields are consistent. Compustat observations are then merged into the monthly CRSP panel using only information that would have been available at the prediction date. To avoid look-ahead bias, we impose a three-month reporting lag after the Compustat data date and discard stale observations beyond the allowed availability window. The resulting dataset is a monthly stock-level panel containing market returns, firm identifiers, lagged accounting information, and factor-state predictors.

2.2 Target Definition and Sample Split

The prediction target is the one-month-ahead stock return, $y_{i,t} = R_{i,t+1}$, where $R_{i,t+1}$ is the realized return of stock i in the month following date t . Observations without a valid consecutive next-month return are removed.

For classification models, labels are constructed month by month by sorting stocks according to their realized next-month returns. We define three classes: bottom quintile, middle three quintiles, and top quintile. This target is aligned with portfolio construction, where the objective is primarily to rank stocks cross-sectionally rather than forecast exact return magnitudes.

We use a chronological split to preserve the time-series structure of the problem. The training set covers 1990–2010, the validation set covers 2011–2015, and the test set covers 2016–2024. Hyperparameters, model choices, score definitions, and portfolio rules are selected using the validation period only. The test period is reserved for final out-of-sample evaluation.

2.3 Predictor Construction

The final feature set combines three predictor blocks. The first block contains return-based variables computed from each stock’s CRSP history: short-term lagged returns, six- and twelve-month momentum measures, rolling volatility, lagged market and excess returns, and rolling beta. All return-based features are computed within each PERMNO time series and shifted so that features available at month t never use the future target return $R_{i,t+1}$.

The second block contains JKP factor-state variables. Since the available JKP data are monthly value-weighted factor returns rather than firm-level characteristics, we treat them as market-wide state variables and merge them by calendar month. We select sixteen interpretable factors and construct three lagged state variables for each: one-month factor return, twelve-month factor momentum, and twelve-month factor volatility. This yields 48 factor-state predictors that allow the models to condition stock-level predictions on the broader asset-pricing environment.

The third block contains Compustat-based accounting variables. Rather than using raw accounting levels directly, we construct transformed and scale-adjusted features related to firm size, profitability, cash-flow quality, investment, financing, and payout behavior. The final accounting block contains 29 selected variables and missingness indicators for variables with substantial missing values. Feature selection is performed on the training period only. Accounting variables are winsorized by month at the 1st and 99th percentiles and imputed using the same-month cross-sectional median, which limits the influence of extreme firm-level ratios while preserving the monthly cross-sectional structure.

2.4 Leakage Prevention

All data processing steps are designed to avoid look-ahead bias. The target is the next-month return, and rows without a valid future return are removed before modeling. Return predictors are lagged within each stock, JKP predictors use lagged factor information, and Compustat data are merged only after a reporting lag. Feature selection, winsorization, imputation, and missingness-indicator construction are performed without using target or label columns. This ensures that the model at month t only uses information that would have been available at the portfolio formation date.

3 Predictive Models

All models are ultimately evaluated through a ranking score $s_{i,t}$ assigned to each stock i at month t . For classification models, we convert class probabilities into a scalar ranking score,

$$s_{i,t} = \hat{p}_{i,t}^{top} - \hat{p}_{i,t}^{bottom}.$$

This corresponds to the expected class value when the bottom, middle, and top classes are encoded as -1 , 0 , and 1 . The resulting score is then evaluated using cross-sectional Rank IC, a standard measure of the association between forecasts and realized returns in active portfolio management [4]. This score is directly aligned with cross-sectional stock selection: high values indicate stocks more likely to be in the top quintile, while low values indicate stocks more likely to be in the bottom quintile.

We use this score during training and model selection because it reduces each model output to a single cross-sectional ranking signal, making regression and classification models directly comparable. During evaluation, however, we also consider alternative scores and portfolio rules because a good statistical ranking does not necessarily imply the best investment performance once concentration, turnover, shorting, and risk are taken into account.

3.1 Benchmark Models

We keep the benchmark set deliberately small. First, we use a naive momentum signal, which ranks stocks according to their past returns and provides a simple economic reference point. Second, we train an XGBoost classifier, a strong nonlinear tabular benchmark based on gradient-boosted decision trees [1]. The XGBoost classifier is evaluated using the common top-minus-bottom probability score defined above. For portfolio evaluation, we also compare strategies to a passive buy-and-hold SPY benchmark.

3.2 Static Deep Learning Models

We next estimate two static deep learning models: a multilayer perceptron (MLP) and an FT-Transformer. Both models use the same monthly tabular feature vector $x_{i,t}$ as the benchmark models. They are called static because each stock-month observation is processed independently; temporal information only enters through the engineered lagged and rolling predictors.

The MLP is a standard feed-forward neural network that maps the standardized feature vector through fully connected layers with nonlinear activations, dropout, and a final prediction head [2]. It serves as a generic neural benchmark for testing whether nonlinear combinations of return, factor, and accounting predictors improve stock ranking.

The FT-Transformer is a tabular architecture based on self-attention [3]. Each numerical predictor is embedded into a feature token, and attention layers learn interactions across features before the final prediction head. This is useful because return predictability may depend on combinations of characteristics rather than on individual predictors in isolation.

Both static models are selected using validation Rank IC rather than training loss, since the objective is to produce a useful cross-sectional ranking signal rather than to minimize prediction error alone.

3.3 Temporal–Tabular Transformer (TTT) Model

The Temporal–Tabular Transformer (TTT) is the main deep learning architecture considered in this project. Its purpose is to combine two types of information: the recent time-series evolution of each stock’s predictors and the contemporaneous cross-sectional feature vector used by the static tabular models.

For each stock i at month t , the temporal branch receives a rolling sequence of the previous L monthly feature vectors,

$$X_{i,t} = (x_{i,t-L+1}, \dots, x_{i,t}).$$

These observations are projected into latent embeddings, augmented with positional information, and processed by Transformer-style self-attention blocks [7]. This produces a temporal representation $h_{i,t}^{\text{temp}}$, which summarizes how the stock’s return dynamics, firm characteristics, and factor-state variables have evolved over time.

In parallel, a tabular branch encodes the current stock-month feature vector $x_{i,t}$, producing a contemporaneous representation $h_{i,t}^{\text{tab}}$. This branch preserves the strength of standard tabular models by focusing on the information available at the portfolio formation date.

The two branches are combined through a learned gated fusion mechanism. After projecting both representations to a common dimension, the model computes

$$g_{i,t} = \sigma \left(W_g [h_{i,t}^{\text{temp}}; h_{i,t}^{\text{tab}}] + b_g \right),$$

and forms the fused representation

$$h_{i,t}^{\text{fused}} = g_{i,t} \odot W_{\text{temp}} h_{i,t}^{\text{temp}} + (1 - g_{i,t}) \odot W_{\text{tab}} h_{i,t}^{\text{tab}},$$

where $\odot$ denotes element-wise multiplication. This allows the model to adaptively weight temporal and contemporaneous information for each stock-month observation.

The temporal, tabular, and fused representations are then passed to a final classification head that predicts the bottom, middle, and top future-return classes. The resulting class probabilities are converted into the same scalar ranking scores as for the other classifier models. This makes the TTT directly

comparable to XGBoost, the MLP, and the FT-Transformer, while testing whether explicit sequence modeling adds predictive value beyond engineered lagged features alone.

In the portfolio results below, TTT refers specifically to the *Small TTT* configuration, which was selected on the validation period. Other TTT configurations are reported only for forecasting comparison. The different configurations are presented in Appendix.

4 Evaluation

4.1 Forecasting Evaluation

Since the objective is cross-sectional portfolio formation, we evaluate models primarily by their ability to rank stocks within each month. For a score $s_{i,t}$, the monthly Rank IC is defined as the Spearman rank correlation between predicted scores and realized next-month returns:

$$\text{RankIC}_t = \rho_S(s_{i,t}, R_{i,t+1}).$$

A positive Rank IC means that stocks assigned higher scores tend to earn higher future returns. We report the average monthly Rank IC,

$$\overline{\text{RankIC}} = \frac{1}{T} \sum_{t=1}^T \text{RankIC}_t,$$

together with the corresponding time-series t -statistic,

$$t = \frac{\overline{\text{RankIC}}}{\text{sd}(\text{RankIC}_t)/\sqrt{T}}.$$

The t -statistic is used as a signal-stability diagnostic rather than as a strict independent hypothesis test, since monthly Rank ICs may be serially correlated.

We also report the IC hit rate, defined as the fraction of months with positive Rank IC, and the realized top-bottom spread, defined as the average future return of the highest-ranked stocks minus that of the lowest-ranked stocks. Rank IC captures statistical ranking quality, while the top-bottom spread gives a direct measure of whether the ranking separates future winners from losers.

4.2 Portfolio Construction

Model scores are converted into monthly portfolios. The main classifier score is

$$s_{i,t}^{\text{cls}} = \hat{p}_{i,t}^{\text{top}} - \hat{p}_{i,t}^{\text{bottom}},$$

where $\hat{p}_{i,t}^{\text{top}}$ and $\hat{p}_{i,t}^{\text{bottom}}$ are the predicted probabilities of the top and bottom future-return classes. We also consider an expected-return score,

$$ER_{i,t}^{\text{train}} = \sum_{k=0}^2 P(y_{i,t+1} = k \mid X_{i,t}) \hat{\mu}_k^{\text{train}},$$

where $\hat{\mu}_k^{\text{train}}$ is the average realized next-month return of class k in the training sample. This maps class probabilities into return units without using validation or test returns.

We evaluate three portfolio rules. The long-only rule buys the highest-ranked stocks. The long-short rule buys the highest-ranked stocks and shorts the lowest-ranked stocks. The 130/30 rule invests 130% in the long leg and 30% in the short leg, giving a net long exposure of 100%. For top and bottom sets L_t and S_t , the basic long-short return is

$$r_{t+1}^{LS} = \frac{1}{|L_t|} \sum_{i \in L_t} R_{i,t+1} - \frac{1}{|S_t|} \sum_{i \in S_t} R_{i,t+1}.$$

Threshold-based variants are also tested. Absolute thresholds are applied to the ER^{train} score, while relative thresholds use the within-month standardized score

$$z_{i,t} = \frac{s_{i,t} - \bar{s}_t}{\sigma_t(s)}.$$

These rules test whether the magnitude of the score contains useful information beyond its ranking.

4.3 Portfolio Evaluation

Portfolio performance is evaluated using monthly net returns. We report annualized return, annualized volatility, Sharpe ratio, maximum drawdown, turnover, and alpha–beta estimates relative to SPY. Annualized return and volatility are computed as $\mu_{\text{ann}} = 12\bar{r}_t$, $\sigma_{\text{ann}} = \sqrt{12}\text{sd}(r_t)$, and the Sharpe ratio is $\text{Sharpe} = \frac{\mu_{\text{ann}}}{\sigma_{\text{ann}}}$.

Transaction costs are applied using a one-way cost of 25 basis points in the final portfolio analysis:

$$r_t^{\text{net}} = r_t^{\text{gross}} - c \cdot \text{Turnover}_t, \quad c = 0.0025.$$

Turnover is computed from changes in portfolio weights and measures trading intensity.

Finally, market-adjusted performance is measured by the regression

$$R_{p,t}^{\text{net}} = \alpha + \beta R_t^{\text{SPY}} + \epsilon_t.$$

SPY is used as a liquid and investable proxy for the U.S. equity market. The resulting alpha and beta help distinguish stock-selection performance from broad market exposure, while remaining only a one-factor benchmark.

5 Results

5.1 Model and Score Selection

Before evaluating the final strategies on the test set, we perform model, score, and portfolio-rule selection on the validation period. The candidate models are the MLP, the FT-Transformer, and the temporal tabular model. For each model, we consider two scoring rules. The first is the classifier score,

$$s_{i,t}^{\text{cls}} = \hat{p}_{i,t}^{\text{top}} - \hat{p}_{i,t}^{\text{bottom}},$$

which ranks stocks according to their predicted probability of belonging to the top rather than the bottom future-return group. The second is an expected-return score, denoted $ER_{i,t}^{\text{train}}$, which maps predicted class probabilities into an expected return using average class returns estimated on the training set.

For each model-score pair, we evaluate a grid of portfolio rules and threshold parameters on the validation set. The portfolio rules are long-only, dollar-neutral long-short, and 130/30. For the expected-return score $ER_{i,t}^{\text{train}}$, we test absolute thresholds $\tau \in \{\text{none}, 0, 0.01, 0.02\}$. For the classifier score, we test relative z-score thresholds $\tau_z \in \{\text{none}, 0, 0.25, 0.5, 1, 1.5\}$. Finally, each candidate is evaluated under transaction-cost assumptions of $c = 25$ bps. This grid allows us to compare not only the models themselves, but also the sensitivity of their signals to score definitions, portfolio construction rules, thresholding, and implementation costs.

The selection procedure is based only on the validation period. We first discard candidates with weak or unstable forecasting and portfolio properties. Specifically, we require positive average Rank IC, a Rank IC t -statistic above 1.96, maximum drawdown above -40% , and average monthly turnover below 1.75. Among the remaining candidates, strategies are ranked by transaction-cost-adjusted Sharpe ratio.

This procedure separates model selection from final evaluation. The validation set is used to choose the model, score, portfolio rule, and threshold, while the test set is reserved for final out-of-sample performance. Some strategies that fail the selection filters still generate high returns or Sharpe ratios, but usually with large drawdowns, high volatility, or concentrated performance in a few years. We therefore treat them as aggressive variants and report them separately in the robustness analysis rather than as main results.

5.2 Forecasting Performance

Table 1 reports the main forecasting metrics for the strongest non-deep-learning benchmark and the core deep learning models. Since the objective is cross-sectional portfolio formation, we focus on ranking rather than pointwise return accuracy. Rank IC measures the monthly Spearman correlation between model scores and next-month returns; IC hit is the fraction of months with positive Rank IC; and the top-bottom spread is the realized return difference between the highest- and lowest-ranked stocks.

Table 1: Forecasting performance of core models

Model	Val. Rank IC	Val. IC hit	Val. T-B spread	Test Rank IC	Test IC hit	Test T-B spread
XGBoost (Baseline)	0.0748***	83.33%	0.0103**	0.0795***	75.70%	0.0112**
MLP	0.0736***	85.00%	0.0091**	0.0957***	78.50%	0.0104*
Small FT-Transformer	0.0936***	78.33%	0.0111**	0.0933***	73.83%	0.0074
Tiny TTT	0.0975***	88.33%	0.0132**	0.1028***	82.24%	0.0094
Small TTT	0.0931***	81.67%	0.0128**	0.0897***	80.37%	0.0093

Notes: T-B spread denotes the average realized return spread between the top- and bottom-ranked stocks. *, **, and *** denote significance at the 10%, 5%, and 1% levels based on the time series of monthly Rank ICs; these should be interpreted as signal-persistence diagnostics rather than strict independent tests.

All models produce positive and statistically significant Rank ICs on both validation and test periods, indicating persistent cross-sectional ranking ability. XGBoost is a strong benchmark, with a test Rank IC of 0.0795 and the largest test top-bottom spread, 1.12% per month. The static neural models improve the ranking metric: the MLP reaches a test Rank IC of 0.0957, while the FT-Transformer remains stable between validation and test but has a weaker realized top-bottom spread.

The temporal tabular transformer models provide the strongest ranking evidence. The Tiny TTT obtains the highest validation and test Rank ICs, 0.0975 and 0.1028, as well as the highest IC hit rates. This suggests that explicitly modeling sequences improves the consistency of stock rankings. However, the best forecasting metric does not necessarily imply the best tradable strategy: portfolio performance also depends on thresholds, turnover, transaction costs, leverage, and drawdowns. The next subsection therefore evaluates whether these signals translate into implementable portfolio returns.

5.3 Portfolio Performance

Portfolio performance is evaluated under a common one-way transaction cost assumption of 25 basis points. This cost regime is used consistently for strategy selection, validation ranking, test evaluation, turnover measurement, and alpha-beta estimation. Monthly portfolio returns are therefore reported net of transaction costs. For each month, gross portfolio returns are computed as

$$R_{p,t}^{gross} = \sum_i w_{i,t} R_{i,t},$$

and transaction costs are applied according to

$$\text{Transaction Cost}_t = \text{Turnover}_t \times \frac{25}{10000}.$$

The net return is therefore

$$R_{p,t}^{net} = R_{p,t}^{gross} - \text{Transaction Cost}_t.$$

All reported returns, Sharpe ratios, drawdowns, alpha estimates, and cumulative performance are based on these net returns. The five highest-ranked validation specifications are reported in Appendix Table 7; all use the Small TTT model, with the best specification using the ER-train score and a gross-normalized long-short rule. For compactness, we label this model as “TTT” in the portfolio tables, but it always refers to the Small TTT configuration.

Table 2 compares the selected TTT strategy with the main benchmarks and model families. The TTT strategy remains the strongest performer on the test set in terms of annualized return and Sharpe ratio, with a net annualized return of 38.8% and a Sharpe ratio of 1.43. This is substantially above the SPY buy-and-hold benchmark, which obtains a test Sharpe of 0.93. The TTT also delivers a large SPY-adjusted alpha of 41.7% on the test period, with a negative SPY beta of -0.30 , indicating that its performance is not mainly compensation for broad market exposure.

However, the main weakness of the TTT strategy is its downside risk. Despite its high average return, it suffers a test maximum drawdown of -62.8% , which is substantially worse than SPY, the FT-Transformer, and the MLP representative strategy. This indicates that the selected TTT portfolio

captures a strong cross-sectional signal, but with severe left-tail risk. The FT-Transformer is less profitable, but it has a more stable risk profile: its test Sharpe is 0.92, close to SPY, with a much smaller maximum drawdown of -25.0% and the highest Rank IC among the reported models. The MLP also has a lower drawdown than TTT, but its test Sharpe falls to 0.79.

The non-deep-learning benchmarks remain useful for context. Naive momentum performs well on validation but collapses on test, with a test Sharpe of only 0.14 and a maximum drawdown of -54.6% . XGBoost is more robust than naive momentum and delivers a positive test Sharpe of 0.84, but it remains below the best deep-learning portfolios. Its turnover is also high, especially on the test period, which makes it more exposed to trading-cost erosion.

Table 2: Portfolio performance comparison at 25 bps

Strategy	Split	Net ret.	Vol.	Sharpe	α_{SPY}	β_{SPY}	MDD	Turnover	Rank IC
B&H SPY	Validation	12.4%	11.7%	1.06	0.0%	1.00	-16.2%	0.00	–
B&H SPY	Test	14.4%	15.5%	0.93	0.0%	1.00	-23.9%	0.00	–
Naive momentum	Validation	18.5%	16.3%	1.14	16.5%	0.16	-12.9%	1.26	0.071 (4.15)
Naive momentum	Test	3.4%	24.4%	0.14	10.9%	-0.29	-54.6%	1.38	0.047 (3.78)
XGBoost	Validation	19.5%	16.5%	1.18	21.4%	-0.17	-15.8%	1.56	0.054 (4.15)
XGBoost	Test	19.5%	23.2%	0.84	26.4%	-0.37	-51.0%	1.77	0.051 (5.86)
FT-Transformer	Validation	12.2%	9.3%	1.31	3.4%	0.69	-10.4%	0.45	0.094 (5.97)
FT-Transformer	Test	10.6%	11.5%	0.92	1.5%	0.61	-25.0%	0.44	0.093 (6.19)
MLP	Validation	13.6%	9.2%	1.49	12.7%	0.05	-5.8%	0.91	0.040 (2.47)
MLP	Test	9.3%	11.8%	0.79	11.4%	-0.12	-31.9%	0.83	0.074 (7.09)
TTT	Validation	71.1%	29.1%	2.44	54.4%	0.38	-27.5%	0.80	0.051 (4.16)
TTT	Test	38.8%	27.0%	1.43	41.7%	-0.30	-62.8%	0.81	0.072 (7.37)

Notes: B&H SPY is the buy-and-hold SPY benchmark and is not a model-based strategy. For model-based strategies, the reported specification is selected using validation data only. Returns, volatilities, and alphas are annualized. Alpha and beta are estimated relative to SPY. MDD denotes maximum drawdown. Rank IC t -statistics are reported in parentheses.

Alpha and beta are estimated from the time-series regression. SPY is used as the market proxy because it is liquid, investable, and provides a standard benchmark for U.S. equity exposure. This choice makes alpha and beta easy to interpret, but it is not a perfect match for the full CRSP-based investment universe. In particular, SPY tracks the S&P 500 and may not capture the behavior of smaller or less liquid stocks included in the prediction panel.

Overall, the portfolio results show a clear trade-off. The TTT strategy has the strongest net performance and remains profitable out of sample, but its drawdown is too large to interpret the strategy as directly deployable without additional risk control. The FT-Transformer provides weaker portfolio returns but a more stable risk profile and the strongest ranking signal, while XGBoost remains a useful non-deep-learning benchmark. The next subsection therefore examines robustness and risk-control issues, including whether the high-return strategies remain attractive once downside risk and model-selection uncertainty are taken into account.

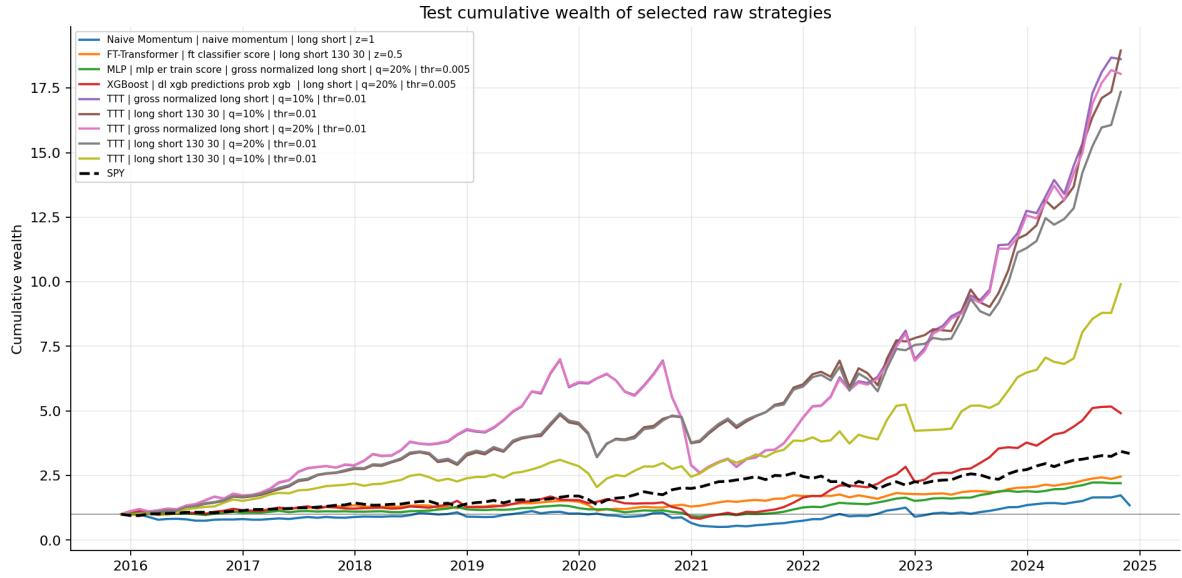


Figure 1: Test-period cumulative wealth of selected raw strategies and the SPY benchmark at 25 bps.

5.4 Robustness Checks

The previous results show that the TTT model produces the strongest portfolio performance among the selected raw strategies, but also that this performance comes with large drawdowns. We therefore run three robustness checks. First, we compare the out-of-sample behavior of the top validation-selected TTT specifications. Second, we inspect the drawdown paths of the selected strategies. Third, we test simple risk-control overlays on the top 300 validation-selected specifications. The overlays are not used in the main performance table; they are reported only as robustness and risk-management diagnostics.

Table 3: Out-of-sample performance of the top validation-selected TTT specifications

Val. rank	Portfolio rule	Net ret.	Vol.	Sharpe	α_{SPY}	β_{SPY}	MDD	Turnover	Gross exp.
1	Gross-normalized L/S, $q = 10\%$, threshold 0.01	38.8%	27.0%	1.43	42.7%	-0.63	-62.8%	0.81	1.00
2	Gross-normalized L/S, $q = 20\%$, threshold 0.01	38.3%	26.9%	1.43	42.3%	-0.62	-62.7%	0.75	1.00
3	130/30 L/S, $q = 10\%$, threshold 0.01	39.1%	22.0%	1.77	30.2%	0.65	-33.8%	0.91	1.60
4	130/30 L/S, $q = 20\%$, threshold 0.01	37.7%	21.7%	1.74	29.0%	0.67	-34.4%	0.75	1.60
5	130/30 L/S, $q = 10\%$, threshold 0.01	29.3%	22.1%	1.33	24.3%	0.47	-34.0%	0.92	1.60

Notes: The table reports test-period performance for the five highest-ranked validation TTT specifications. All returns, volatilities, and alphas are annualized. Alpha and beta are estimated relative to SPY. The ranking is fixed using validation-period net Sharpe only. Gross exposure is the average absolute portfolio exposure.

Table 3 shows that the TTT signal is not tied to a single portfolio rule. The two gross-normalized long-short variants have the highest validation ranks and remain highly profitable on test, but they suffer very large maximum drawdowns of about -63% . By contrast, the 130/30 variants obtain comparable or even higher test Sharpe ratios, while reducing maximum drawdown to about -34% . This suggests that the TTT score contains a robust cross-sectional signal, but that portfolio construction strongly affects the downside profile. The best validation-Sharpe implementation is therefore not necessarily the most attractive out-of-sample implementation once drawdown risk is considered.

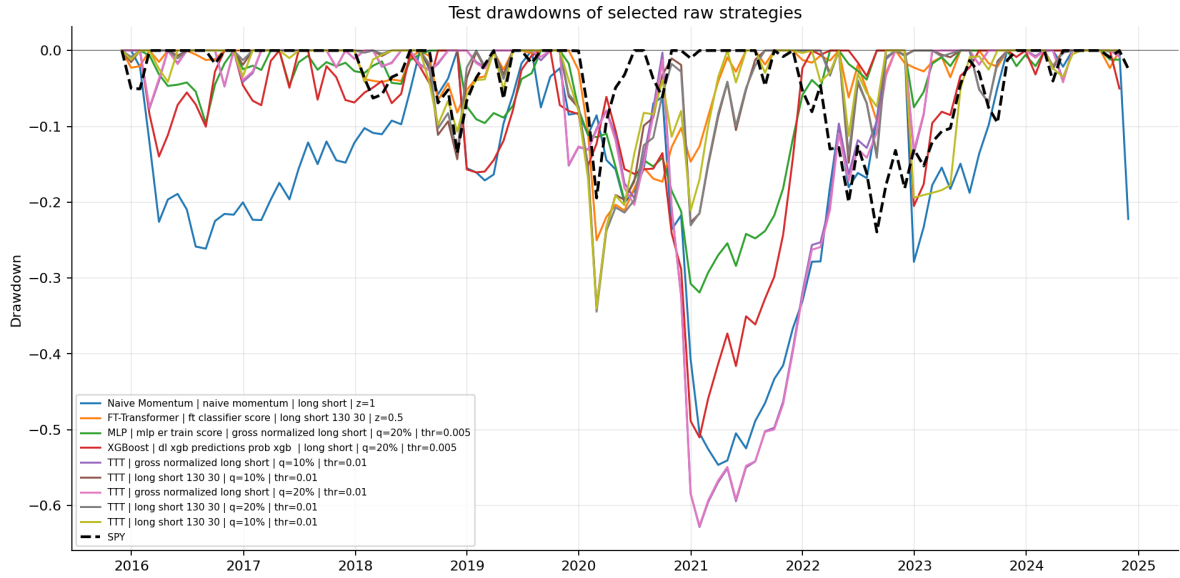


Figure 2: Test-period drawdowns of selected raw strategies and the SPY benchmark at 25 bps.

Figure 2 confirms this point. The high-return TTT strategies generate strong cumulative performance, but the drawdown paths are much less stable than the headline Sharpe ratios suggest. The largest losses are concentrated around the 2020–2022 period, where several long-short strategies experience sharp losses before recovering. This motivates a separate check of whether simple exposure-control rules can reduce drawdown without changing the underlying predictive model.

We test two lagged risk-control overlays on the top 300 validation-selected specifications. The first classifies the strategy’s own wealth path into drawdown states and reduces exposure during shock or grinding drawdowns. The second combines this drawdown-state rule with a volatility target, further reducing exposure when recent realized strategy volatility is high. Both rules use only information available through month t and apply the exposure multiplier to month $t + 1$. Full details are provided in Appendix C.

Table 4: Average effect of risk-control overlays on the top 300 specifications

Split	Overlay	Δ ret.	Δ vol.	Δ Sharpe	Δ MDD	Share MDD improved	Avg. λ
Validation	DD speed/phase	-2.26 pp	-1.74 pp	-0.006	+3.44 pp	96.7%	0.91
Validation	DD speed/phase + vol target	-4.00 pp	-2.14 pp	-0.127	+3.48 pp	98.0%	0.85
Test	DD speed/phase	-2.66 pp	-2.91 pp	-0.016	+9.55 pp	99.3%	0.88
Test	DD speed/phase + vol target	-3.27 pp	-3.75 pp	-0.003	+13.03 pp	99.7%	0.83

Notes: The table reports average changes relative to the raw strategy. Positive Δ MDD means that maximum drawdown becomes less negative. Average λ is the mean exposure multiplier applied by the overlay.

Table 4 shows that the overlays consistently reduce risk, but at the cost of lower returns. On the test period, the drawdown speed/phase overlay improves average maximum drawdown by 9.55 percentage points, while the hybrid drawdown-plus-volatility overlay improves it by 13.03 percentage points. These improvements are highly systematic: maximum drawdown improves for more than 99% of the top 300 specifications on the test set.

However, the validation results are more mixed. Both overlays reduce volatility and maximum drawdown, but they do not improve average validation Sharpe. The hybrid overlay is especially conservative on validation, reducing annualized return by about 4.0 percentage points and lowering the average Sharpe. This is why we do not include overlays in the main performance comparison. They are useful risk-management extensions, but selecting them as part of the final strategy would add another layer of tuning and model-selection risk.

Overall, the robustness checks lead to two conclusions. First, the TTT ranking signal survives across several portfolio constructions, but the choice of portfolio rule has a large impact on drawdown.

Second, simple lagged risk-control overlays can substantially reduce maximum drawdown, especially out of sample, but they trade off return and introduce an additional design choice. We therefore treat them as promising robustness evidence rather than as the main reported strategy.

5.5 Limitations

Several limitations remain. First, the project involves substantial specification search. We compare many model architectures, scores, gates, thresholds, and portfolio rules. Although final performance is evaluated on a held-out test set, the validation period is used repeatedly to guide modelling and strategy design. The best validation-performing strategies may therefore partly reflect selection noise, especially because equity-return prediction has a low signal-to-noise ratio.

Second, the deep learning models improve flexibility but reduce interpretability. The FT-Transformer and the Temporal-Tabular Transformer can capture nonlinear interactions between return dynamics, firm characteristics, and factor states, but it is difficult to identify the exact economic mechanism behind their predictions. This limits how easily the strategy can be monitored or diagnosed during periods of underperformance.

Third, the trading assumptions remain simplified. We impose a constant 25 bps one-way transaction cost, but do not model time-varying spreads, market impact, short-borrow fees, financing costs, short-sale constraints, or capacity limits. These frictions are particularly important for long-short and high-turnover strategies, and could materially reduce realized performance.

Fourth, alpha and beta are estimated relative to SPY. This provides a liquid and interpretable market benchmark, but SPY tracks the S&P 500 rather than the full stock universe used in the portfolio construction. A low or negative SPY beta therefore does not rule out exposure to other systematic factors such as size, value, momentum, liquidity, or sector risk.

Finally, the evaluation relies on a single chronological split. The test period includes several regimes, but it remains one historical path. A stronger validation design would use rolling or expanding-window retraining, multiple out-of-sample blocks, and factor-risk attribution. Overall, the results should be interpreted as evidence of economically useful ranking information, not as a directly deployable "safe" trading strategy.

6 Conclusion

This project studies whether machine learning models can extract useful cross-sectional signals for one-month-ahead stock return prediction and convert them into portfolio performance. The results show that the models do learn economically meaningful rankings. All main models achieve positive and statistically significant Rank ICs on both validation and test periods, and the temporal-tabular architectures obtain the strongest forecasting results. This supports the idea that combining firm characteristics, factor-state variables, and lagged return dynamics can improve cross-sectional stock ranking.

The portfolio results are also economically meaningful. After 25 bps one-way transaction costs, the selected TTT strategy outperforms the SPY benchmark and the non-deep-learning baselines in terms of test annualized return and Sharpe ratio. The result is not only statistical: the model signal translates into profitable long-short portfolio construction out of sample.

However, the main result is not that the strategy is immediately deployable. The strongest validation-selected TTT implementation suffers a large test maximum drawdown, showing that high Rank IC and high Sharpe do not eliminate path risk. Robustness checks show that alternative TTT portfolio rules, especially 130/30 constructions, can preserve strong returns while materially reducing drawdowns. Risk-control overlays further reduce maximum drawdown across a broad set of specifications, but they also lower returns and introduce an additional layer of design choices.

Overall, the main contribution of the project is an end-to-end empirical pipeline: linked financial data, leakage-controlled features, machine learning ranking models, validation-based portfolio selection, and transaction-cost-adjusted performance evaluation. The results suggest that deep learning, especially temporal-tabular modeling, can improve cross-sectional stock selection. At the same time, the strategy remains exposed to specification risk, implementation frictions, and drawdown risk. A production-level extension would require rolling retraining, stronger factor-risk attribution, more realistic transaction-cost modeling, and formally validated risk controls.

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A Additional Implementation Details

A.1 Predictor Blocks

Table 5 summarizes the final predictor set used by the models. The main text describes the construction logic; this table reports the resulting feature blocks.

Table 5: Summary of predictor blocks

Block	No.	Description
Return features	11	Lagged returns, momentum, volatility, lagged market return, excess return, and beta.
JKP factor states	48	One-month return, twelve-month momentum, and twelve-month volatility for 16 factor portfolios.
Accounting features	29	Transformed and scale-adjusted variables covering size, profitability, cash flow, investment, financing, and payout.
Missingness indicators	10	Indicators for accounting variables with substantial missingness.

A.2 Deep Learning Architecture Configurations

Table 6 reports the main architecture configurations used in the deep learning experiments. The MLP is used as a static neural benchmark, the FT-Transformer is used as an attention-based tabular model, and the TTT variants combine a temporal branch with a contemporaneous tabular branch.

Table 6: Deep learning architecture configurations

Model	Input	Hidden / token dim.	Layers	Heads	Fusion dim.	Output
MLP	Tabular	128–64–32	3	–	–	3 classes
FTT-tiny	Tabular	16	1	2	–	3 classes
FTT-small	Tabular	32	2	4	–	3 classes
TTT-tiny	Temporal + tabular	16 / 8	1 + 1	1 + 1	64	3 classes
TTT-small	Temporal + tabular	32 / 16	1 + 1	2 + 2	128	3 classes
TTT-default	Temporal + tabular	64 / 32	3 + 2	4 + 4	256	3 classes

The retained TTT configuration in the final results is the default model. It has higher capacity than the tiny and small presets, allowing richer nonlinear interactions between lagged stock dynamics and current tabular characteristics. This also increases overfitting risk, which is why validation Rank IC and out-of-sample portfolio performance are both used in the model assessment.

B Additional Portfolio Results

B.1 Top Validation-Selected Specifications

Table 7 reports the five highest-ranked validation strategies at 25 bps transaction costs. These specifications are selected using validation net Sharpe after imposing the forecasting and implementation filters described in the main text.

Table 7: Top validation-selected portfolio specifications at 25 bps

Rank	Model	Portfolio rule	Net ret.	Vol.	Sharpe	α_{SPY}	β_{SPY}	MDD	Turnover	Rank IC
1	TTT	Gross-normalized L/S, $q = 10\%$, threshold 0.01	71.1%	29.1%	2.44	54.4%	0.38	-27.5%	0.80	0.051 (4.16)
2	TTT	Gross-normalized L/S, $q = 20\%$, threshold 0.01	70.5%	29.0%	2.43	54.1%	0.38	-27.3%	0.75	0.051 (4.16)
3	TTT	130/30 L/S, $q = 10\%$, threshold 0.01	55.6%	23.2%	2.40	30.3%	1.39	-18.6%	0.92	0.051 (4.16)
4	TTT	130/30 L/S, $q = 20\%$, threshold 0.01	54.0%	22.9%	2.36	29.3%	1.38	-18.3%	0.77	0.051 (4.16)
5	TTT	130/30 L/S, $q = 10\%$, threshold 0.01	36.8%	16.2%	2.28	27.1%	0.48	-13.6%	1.01	0.058 (5.17)

Notes: Specifications are ranked by validation net Sharpe after requiring positive Rank IC, Rank IC t -statistic above 1.96, maximum drawdown above -40% , and average turnover below 1.75. Returns, volatilities, and alphas are annualized. Alpha and beta are estimated relative to SPY. Rank IC t -statistics are reported in parentheses.

The top validation specifications are all TTT variants. The best two use the same expected-return score and gross-normalized long-short construction, differing only in the quantile cutoff. The 130/30 alternatives have slightly lower validation returns but materially lower validation drawdowns, which motivates the portfolio-rule robustness analysis in the main text.

B.2 Annual Return and Drawdown Breakdown

Figure 3 reports annual net returns and within-year maximum drawdowns for two TTT-based strategies: the main validation-selected strategy and the risk-balanced 130/30 alternative. Both are evaluated under the same 25 bps transaction-cost assumption.

Annual Net Return and Within-Year Max Drawdown at 25 bps

Period	Main return	Main MDD	Risk-balanced return	Risk-balanced MDD
Validation 2010	-1.3%	-1.3%	0.2%	0.0%
Validation 2011	48.9%	-3.9%	28.1%	-18.6%
Validation 2012	205.6%	-2.7%	139.1%	-6.1%
Validation 2013	14.5%	-27.5%	42.7%	-12.6%
Validation 2014	95.6%	-5.9%	61.0%	-6.2%
Validation 2015	45.9%	-9.9%	29.3%	-6.9%
Validation AVG	68.2%	-8.5%	50.1%	-8.4%
Test 2015	11.8%	0.0%	-1.5%	-1.5%
Test 2016	53.9%	-7.7%	67.8%	-1.9%
Test 2017	67.6%	-2.2%	66.7%	0.0%
Test 2018	47.7%	-2.6%	18.9%	-14.3%
Test 2019	42.8%	-15.2%	36.9%	-7.6%
Test 2020	-52.4%	-58.4%	-16.3%	-28.4%
Test 2021	63.9%	-10.5%	60.8%	-6.6%
Test 2022	47.5%	-13.6%	29.7%	-14.8%
Test 2023	82.0%	-1.9%	51.3%	-6.9%
Test 2024	46.0%	-3.9%	60.2%	-2.4%
Test AVG	41.1%	-11.6%	37.4%	-8.4%

Figure 3: Annual net return and within-year maximum drawdown at 25 bps for the main TTT strategy and the risk-balanced TTT alternative.

The heatmap shows that the main TTT strategy has very high average annual returns, but its performance is uneven across years. The largest stress episode occurs in 2020, when the main strategy loses 52.4% and reaches a within-year maximum drawdown of -58.4% . The risk-balanced 130/30 strategy reduces the severity of this episode: the 2020 annual loss falls to -16.3% , and the within-year drawdown improves to -28.4% . This confirms that the TTT signal is strong, but that portfolio construction has a major effect on path risk.

C Risk-Control Overlay Methodology

The risk-control overlays are used only in the robustness checks. They are not part of the main portfolio-performance tables. Their purpose is to test whether drawdown risk can be reduced after portfolio construction without changing the underlying prediction model or stock-ranking score.

Let $R_{p,t}^{base}$ denote the monthly net return of the raw strategy before the overlay, and let W_t^{base} denote its cumulative wealth:

$$W_t^{base} = W_{t-1}^{base} (1 + R_{p,t}^{base}).$$

The base-strategy drawdown is

$$DD_t = \frac{W_t^{base}}{\max_{\tau \leq t} W_\tau^{base}} - 1.$$

The overlay assigns an exposure multiplier λ_t , and the scaled strategy return is

$$R_{p,t+1}^{overlay} = \lambda_t R_{p,t+1}^{base}.$$

The multiplier λ_t is computed using only information available through the end of month t and is applied to month $t + 1$. This timing prevents same-month look-ahead.

C.1 Drawdown Speed/Phase Overlay

The first overlay classifies the lagged wealth path into drawdown states. The rule distinguishes between normal periods, shallow drawdowns, sharp drawdowns, persistent drawdowns, and recovery phases. The exposure multiplier follows a monotone risk schedule:

$$\lambda_t^{DD} = \begin{cases} 1.00, & \text{new high or normal state,} \\ 0.85, & \text{shallow drawdown,} \\ 0.75, & \text{recovery or stale recovery,} \\ 0.50, & \text{shock or grinding drawdown.} \end{cases}$$

The intuition is to remain fully invested when the strategy’s own wealth process is healthy, but to reduce exposure during sharp or persistent losses. This is similar in spirit to trend-following and portfolio-insurance rules, but applied to the strategy wealth curve rather than to a market index.

C.2 Drawdown-Phase Volatility-Target Overlay

The second overlay combines the drawdown-state multiplier with a volatility target. Let $\hat{\sigma}_{p,t}$ be recent realized volatility estimated from lagged base-strategy returns. The final multiplier is

$$\lambda_t^{hybrid} = \min \left(\lambda_t^{DD}, \frac{\sigma_t^*}{\hat{\sigma}_{p,t}} \right),$$

where σ_t^* is a state-dependent volatility target, with lower targets in adverse drawdown states. This hybrid rule reduces exposure both when the strategy is in an unfavorable drawdown phase and when recent realized volatility is high relative to the current risk budget.

Both overlays add an additional timing layer. For this reason, they are reported as robustness checks rather than included in the main selected strategy.

D External Libraries and Tools

The project was implemented in Python. Data cleaning, merging, feature construction, and panel manipulation relied primarily on pandas, NumPy, and PyArrow. Statistical evaluation used SciPy and statsmodels. The baseline models and preprocessing used scikit-learn, while XGBoost was used for gradient boosting. The deep learning models were implemented in PyTorch. Matplotlib and seaborn were used for figures, and yfinance was used to retrieve the SPY benchmark.